

Backend Schema & Database Design v2.0

CloudWise AI

Version: 2.0

Status: Active Development

Last Updated: June 2026

1. Purpose

This document defines the database architecture, entity relationships, table structures, indexes, constraints, and future expansion strategy for CloudWise AI.

The database serves as the central source of truth for:

- Users
 - Cloud Accounts
 - Resource Inventory
 - Metrics
 - Recommendations
 - Forecasts
 - Reports
 - AI Context
-

2. Database Technology

Development:

SQLite

Production:

PostgreSQL

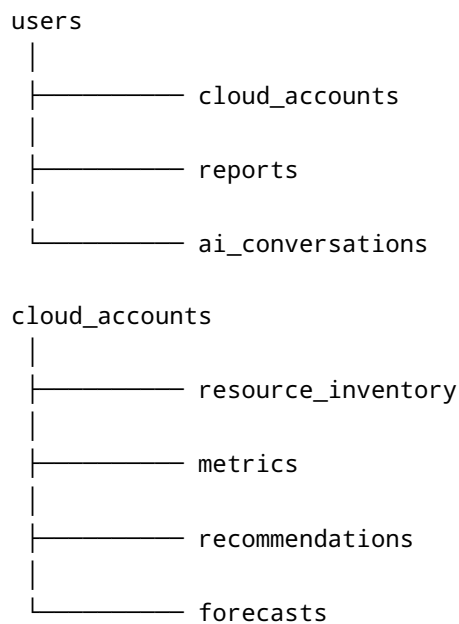
ORM:

SQLAlchemy

Migration Tool:

Alembic

3. Entity Relationship Overview



4. Users Table

Purpose:

Store application users.

Table:

`users`

Fields:

id	UUID PRIMARY KEY
email	VARCHAR(255) UNIQUE NOT NULL
full_name	VARCHAR(255)
hashed_password	TEXT NOT NULL
is_active	BOOLEAN DEFAULT TRUE
is_verified	BOOLEAN DEFAULT FALSE
created_at	TIMESTAMP
updated_at	TIMESTAMP

Indexes:

INDEX(email)

5. Password Reset Tokens

Purpose:

Support forgot password functionality.

Table:

password_reset_tokens

Fields:

id	UUID PRIMARY KEY
user_id	UUID
token	TEXT UNIQUE
expires_at	TIMESTAMP

used	BOOLEAN DEFAULT FALSE
created_at	TIMESTAMP

Relationships:

user
↓
many reset tokens

6. Cloud Accounts

Purpose:

Store AWS and LocalStack connections.

Table:

cloud_accounts

Fields:

id	UUID PRIMARY KEY
user_id	UUID
provider	VARCHAR(50)
account_id	VARCHAR(100)
account_name	VARCHAR(255)
region	VARCHAR(50)
use_localstack	BOOLEAN DEFAULT FALSE
created_at	TIMESTAMP
updated_at	TIMESTAMP

Example:

AWS Account

or

LocalStack Environment

Indexes:

`INDEX(user_id)`

`INDEX(account_id)`

7. Resource Inventory

Purpose:

Store discovered cloud resources.

Table:

`resource_inventory`

Fields:

<code>id</code>	<code>UUID PRIMARY KEY</code>
<code>cloud_account_id</code>	<code>UUID</code>
<code>resource_id</code>	<code>VARCHAR(255)</code>
<code>resource_name</code>	<code>VARCHAR(255)</code>
<code>resource_type</code>	<code>VARCHAR(50)</code>
<code>region</code>	<code>VARCHAR(50)</code>
<code>status</code>	<code>VARCHAR(50)</code>
<code>arn</code>	<code>TEXT</code>

tags	JSON
metadata_json	JSON
created_at	TIMESTAMP
updated_at	TIMESTAMP

Supported Types:

EC2
EBS

Future:

S3
RDS
Lambda

Indexes:

INDEX(resource_id)

INDEX(resource_type)

INDEX(region)

INDEX(status)

8. Metrics Table

Purpose:

Store utilization metrics.

Table:

metrics

Fields:

id	UUID PRIMARY KEY
resource_id	UUID
metric_name	VARCHAR(100)
metric_value	FLOAT
timestamp	TIMESTAMP

Examples:

```
CPUUtilization  
  
DiskUsage  
  
MemoryUsage
```

Indexes:

```
INDEX(resource_id)  
  
INDEX(metric_name)
```

9. Cost Records

Purpose:

Store cloud cost information.

Table:

```
cost_records
```

Fields:

id	UUID PRIMARY KEY
cloud_account_id	UUID
service_name	VARCHAR(100)
cost_amount	DECIMAL(10,2)
currency	VARCHAR(10)
cost_date	DATE
created_at	TIMESTAMP

Examples:

EC2
EBS
S3
RDS

Indexes:

INDEX(cost_date)
INDEX(service_name)

10. Recommendations

Purpose:

Store optimization recommendations.

Table:

recommendations

Fields:

id	UUID PRIMARY KEY
resource_id	UUID
recommendation_type	VARCHAR(100)
priority	VARCHAR(20)
estimated_savings	DECIMAL(10,2)
description	TEXT
status	VARCHAR(50)
created_at	TIMESTAMP

Priority Values:

LOW
MEDIUM
HIGH

Status Values:

OPEN
DISMISSED
IMPLEMENTED

Indexes:

INDEX(priority)
INDEX(status)

11. Forecasts

Purpose:

Store cost forecasting results.

Table:

forecasts

Fields:

id	UUID PRIMARY KEY
cloud_account_id	UUID
forecast_period	INTEGER
predicted_cost	DECIMAL(10,2)
confidence_score	FLOAT
generated_at	TIMESTAMP

Examples:

7 Day Forecast

30 Day Forecast

90 Day Forecast

12. Reports

Purpose:

Store generated reports.

Table:

reports

Fields:

id	UUID PRIMARY KEY
user_id	UUID
report_type	VARCHAR(100)
file_path	TEXT
generated_at	TIMESTAMP

Report Types:

Executive Summary

Cost Analysis

Optimization Report

13. AI Conversations

Purpose:

Store AI Copilot interactions.

Table:

ai_conversations

Fields:

id	UUID PRIMARY KEY
user_id	UUID
question	TEXT

response	TEXT
created_at	TIMESTAMP

Purpose:

- Conversation History
- Context Building
- Future Fine-Tuning

14. Relationships

```
users
├── cloud_accounts
├── reports
└── ai_conversations

cloud_accounts
├── resource_inventory
├── forecasts
├── recommendations
└── cost_records

resource_inventory
├── metrics
└── recommendations
```

15. Dashboard Query Sources

Total Resources

Source:

```
resource_inventory
```

Monthly Spend

Source:

```
cost_records
```

Potential Savings

Source:

```
recommendations
```

Health Score

Source:

```
resource_inventory  
metrics  
recommendations
```

16. LocalStack Support

Cloud Account Example:

```
provider = AWS  
  
use_localstack = true
```

Behavior:

```
Frontend Toggle  
↓  
Backend Endpoint Switch  
↓  
LocalStack Resource Discovery  
↓  
Database Storage  
↓  
Dashboard Analytics
```

17. Future Expansion

Version 3.0

Additional Resource Types:

```
S3  
RDS  
Lambda  
CloudFront  
ECS  
EKS
```

Version 4.0

Multi-Cloud:

```
AWS  
  
Azure
```

Google Cloud

18. Data Retention

Metrics:

90 Days

Forecasts:

180 Days

Reports:

365 Days

Audit Logs:

365 Days

Approval

Document Status:

Approved

Version:

2.0